

PYTHON LAB MANUAL - TEST CASES

PROGRAM 1 — Arithmetic Operations

Test Case ID	Input	Expected Output
TC1	10,5	Addition = 15
TC2	20,4	Division = 5
TC3	9,0	Division by zero error

PROGRAM 2 — Even or Odd

Test Case ID	Input	Expected Output
TC1	10	Even
TC2	7	Odd
TC3	-4	Even

PROGRAM 3 — Fibonacci Series

Test Case ID	Input n	Expected Output
TC1	5	0 1 1 2 3
TC2	1	0
TC3	0	No output

PROGRAM 4 — Vowel & Consonant Counter

Test Case ID	Input String	Expected Result
TC1	Hello	Vowels=2, Consonants=3
TC2	AEIOU	Vowels=5
TC3	Python	Vowels=1

PROGRAM 5 — Student Dictionary

Test Case ID	Operation	Expected Output
TC1	Add student	Student stored
TC2	Search roll no	Details displayed
TC3	Invalid roll no	Not found

PROGRAM 6 — Break Continue Pass

Test Case ID	Statement	Expected Result
TC1	break	Loop terminates
TC2	continue	Skips iteration
TC3	pass	No effect

PROGRAM 7 — NumPy Array Operations

Test Case ID	Operation	Expected Output
TC1	Array addition	Element-wise sum
TC2	Array subtraction	Difference
TC3	Reshape array	2x5 matrix

PROGRAM 8 — Pandas Series

Test Case ID	Operation	Expected Output
TC1	Indexing	Correct value
TC2	Slicing	Subset displayed
TC3	Query	Filtered values

PROGRAM 9 — DataFrame Sorting

Test Case ID	Operation	Expected Output
TC1	Sort descending	Highest first
TC2	Filter >=80	Matching rows
TC3	Duplicate values	Sorted correctly

PROGRAM 10 — Missing Value Handling

Test Case ID	Input	Expected Output
TC1	NaN values	Filled with mean
TC2	Median fill	Filled correctly
TC3	No NaN	Same dataset

PROGRAM 11 — Character Occurrence

Test Case ID	Input	Expected Output
TC1	WELCOME	Character frequencies
TC2	aaa	a=3
TC3	Empty string	Empty dictionary

PROGRAM 12 — List CRUD Operations

Test Case ID	Operation	Expected Result
TC1	Insert	Added successfully
TC2	Delete	Removed successfully
TC3	Update	Value updated

PROGRAM 13 — Employee Dictionary CRUD

Test Case ID	Operation	Expected Output
TC1	Create employee	Dictionary created
TC2	Update employee	Modified details
TC3	Delete field	Field removed

PROGRAM 14 — Prime Generator

Test Case ID	Input	Expected Output
TC1	10	2 3 5 7
TC2	2	2
TC3	1	No primes

PROGRAM 15 — Recursive Factorial

Test Case ID	Input	Expected Output
TC1	5	120
TC2	0	1
TC3	-1	Invalid input

PROGRAM 16 — File Handling

Test Case ID	Operation	Expected Output
TC1	Write file	Data stored

TC2	Read file	Contents displayed
TC3	Missing file	File error

PROGRAM 17 — Matplotlib Graphs

Test Case ID	Graph Type	Expected Output
TC1	Line graph	Graph displayed
TC2	Bar graph	Bar chart shown
TC3	Empty dataset	Blank graph

PROGRAM 18 — CSV Data Analysis

Test Case ID	Operation	Expected Output
TC1	Read CSV	Data loaded
TC2	Describe()	Statistics shown
TC3	Value counts	Frequency displayed

PROGRAM 19 — GroupBy Statistics

Test Case ID	Operation	Expected Output
TC1	Group by category	Summary table
TC2	Mean calculation	Correct mean
TC3	Sum calculation	Correct total

PROGRAM 20 — Scikit-learn Dataset

Test Case ID	Operation	Expected Output
TC1	Load iris dataset	Dataset loaded
TC2	Display shape	(150,4)
TC3	Feature names	Displayed

Additional Viva Questions

1. Difference between List and Tuple
2. What is NumPy?
3. Difference between Series and DataFrame
4. What is recursion?
5. What is a generator function?
6. Difference between break and continue
7. Applications of Pandas
8. Why use Matplotlib?
9. What is machine learning dataset?
10. Explain CRUD operations in Python